

Vision HCI Assessment

(CLO-3)

BSSE23B

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Human Computer Interaction

SE305T (Spring-26)

Individual proctored Assessment

LT2-BlockB

Information Technology University (ITU) – Lahore

6-May-2026

Time= 4:00pm - 5:30pm

Total Marks=40

Task-1: CHALLENGE A — BLINK DETECTION (15 marks)

Build a complete blink detection system from scratch.

Your Program Must:

1. Import OpenCV, MediaPipe, NumPy
2. Open webcam using `cv2.VideoCapture(0)`
3. Use MediaPipe FaceMesh to detect 468 face landmarks
4. Compute Eye Aspect Ratio (EAR) using landmarks [362, 385, 387, 263, 373, 380] (left eye) and [33, 160, 158, 133, 153, 144] (right eye)
1. Detect blink transitions: when BOTH eyes close ($EAR < 0.25$) then open ($EAR \geq 0.25$), increment blink counter
2. Display 'Blinks: X' on the frame in real time
3. Press ESC to exit cleanly (close camera, destroy windows)

Acceptance Criteria:

- Code runs without errors
- Blink counter increments only on complete open→close→open cycle
- Counter visible on frame in readable font size
- ESC key exits gracefully

Task-3: CHALLENGE C — GESTURE RECOGNITION (25 marks)

Build a hand gesture recognition system from scratch.

Your Program Must:

4. Import OpenCV, MediaPipe, NumPy
5. Open webcam using `cv2.VideoCapture(0)`
6. Use MediaPipe Hands to detect up to 2 hands (21 landmarks each)
7. For each hand, count extended fingers:
 - a) Compare TIP vs PIP joint y-coordinates for each finger ($TIP.y < PIP.y$ = extended)
 - b) Handle thumb separately using x-coordinate + handedness (Left vs Right hand differs)
 - c) Return count 0–5 per hand
8. Classify gestures based on finger count:
 - a) (0, False) = Fist 🖐️ | (1, False) = One 🖐️ | (2, False) = Peace 🖐️
 - b) (5, False) = Open Hand 🖐️ | (1, True) = Thumbs Up 👍 | (5, True) = High Five 🙌
9. Display finger count + gesture name + emoji on frame for each hand
10. Update every frame
11. Press ESC to exit cleanly

Acceptance Criteria:

- Code runs without errors
- Finger count 0–5 accurate for both hands
- Thumb detection handedness-aware
- Gesture labels correct (use dictionary mapping)
- Display shows both hands if visible
- Output updates smoothly every frame

Submission Details:

- Deadline: Wednesday, May 6, 2026, upto 5:30PM.
- You must submit code files:
 - `hci_assessment_YourName.py` — your complete, working Python code
- Submit using shared Google Form Link in Classroom within the given time.